

ABAP Unit 3 – Exercise 9

Create and Manage Instances

Class: ZCL_9309_LOCAL_CLASS

Package: ZS4D400_9309

1. Objective

Create objects (instances) of the local class LCL_CONNECTION, assign values to their instance attributes, and store three object references in an internal table.

2. Task 1 – Template

Exercise 8 was already completed, so the existing global class ZCL_9309_LOCAL_CLASS was reused. No template copy was required.

3. Task 2 – Create an Instance

Declare a reference variable, create an object with NEW #(), and set its attributes.

```
DATA connection TYPE REF TO lcl_connection.
```

```
connection = NEW #( ).  
connection->carrier_id = 'LH'.  
connection->connection_id = '0400'.
```

4. Task 3 – Manage Several Instances

Declare an internal table whose line type is a reference to LCL_CONNECTION. Reuse CONNECTION to create three separate objects and append each reference.

```
DATA connection TYPE REF TO lcl_connection.  
DATA connections TYPE TABLE OF REF TO lcl_connection.
```

```
connection = NEW #( ).  
connection->carrier_id = 'LH'.  
connection->connection_id = '0400'.  
APPEND connection TO connections.
```

```
connection = NEW #( ).  
connection->carrier_id = 'AA'.  
connection->connection_id = '0017'.  
APPEND connection TO connections.
```

```
connection = NEW #( ).  
connection->carrier_id = 'SQ'.  
connection->connection_id = '0001'.  
APPEND connection TO connections.
```

5. Instance Data

Instance	Carrier ID	Connection ID	Stored
1	LH	0400	Yes
2	AA	0017	Yes
3	SQ	0001	Yes

6. Debugger Result

After the third APPEND, the Variables view showed CONNECTIONS [3x1(8)] Standard Table. This confirms that three object references were stored.

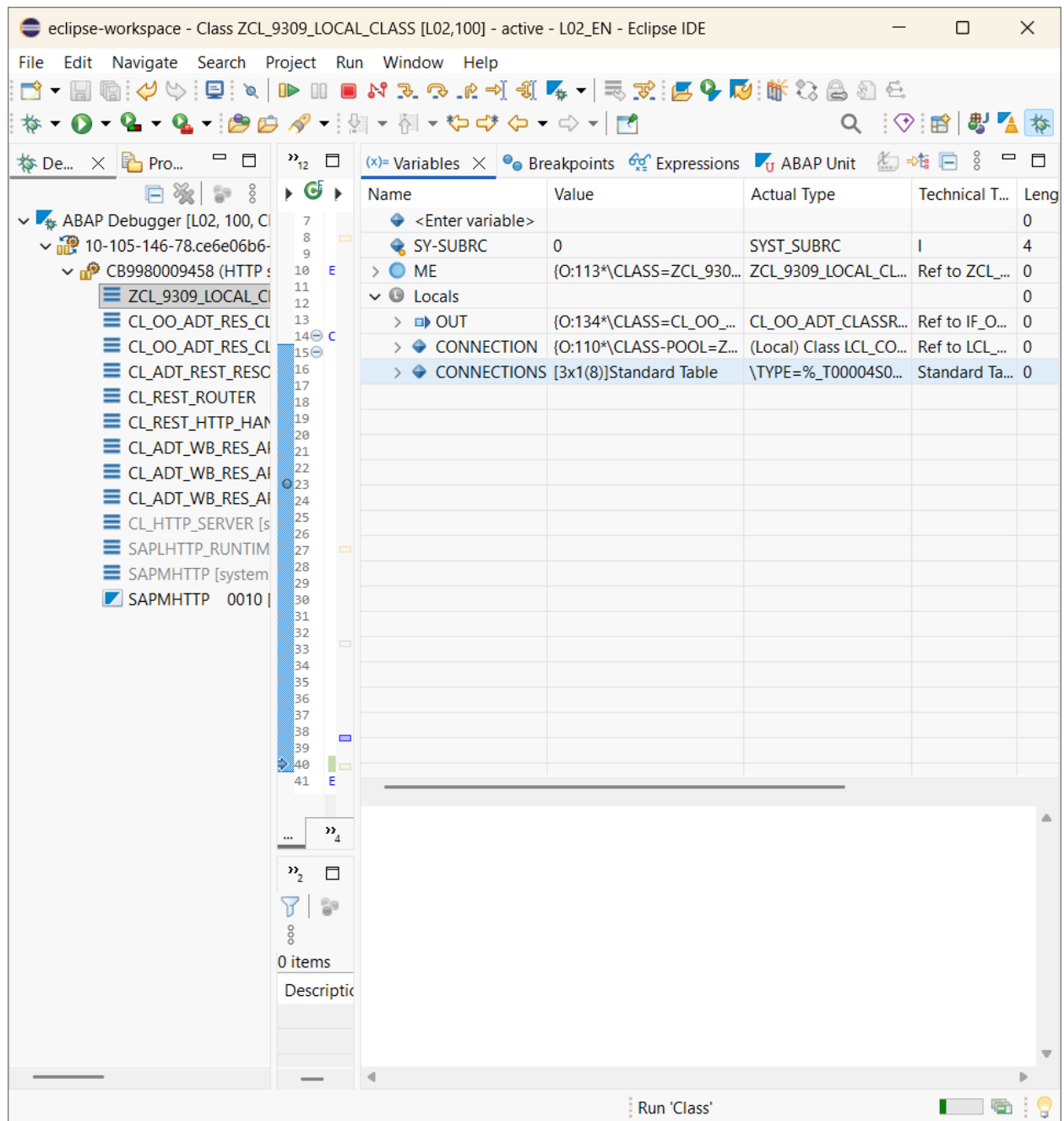


Figure 1 – Debugger showing CONNECTIONS [3x1(8)]

7. Debugging Steps

1. Press Ctrl + F3 to activate the class.
2. Set a breakpoint on an APPEND CONNECTION TO CONNECTIONS statement.
3. Press F9 to run the class.
4. Press F5 to execute one statement at a time.
5. Expand CONNECTIONS in the Variables view.
6. After the third APPEND, verify CONNECTIONS contains 3 entries.

8. Key Concepts

- REF TO creates a reference variable for an object.
- NEW #() creates a new instance.
- The -> operator accesses instance attributes.
- An internal table can store object references.
- The same reference variable can point to different objects after each NEW #().

9. Final Status

COMPLETED – Three LCL_CONNECTION instances were created and stored in CONNECTIONS.